

Supplementary Appendix

Social Vulnerability and Progression to Vision-Threatening Diabetic Retinopathy: A County-Level Analysis of 3,119 United States Counties, 2021

This appendix documents every hypothesis we tested, including those that did not support the analysis reported in the main text. Each is presented as: what is tested conceptually, how we tested it, what we found, and how we interpret it.

Unless stated otherwise, the outcome is the vision-threatening share of diagnosed diabetic retinopathy (VTDR cases divided by all-stage DR cases among adults with diabetes), models are quasibinomial with DR cases as trials, and continuous predictors are standardized so coefficients are per standard deviation on the log-odds scale.

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Test 1. Post-stratification artifact in county surveillance estimates

(1) What is tested. County-level estimates from the Vision and Eye Health Surveillance System (VEHSS) are not direct measurements. They are constructed by computing national prevalence by age, sex, and race, applying those national rates to each county’s demographic composition, and then adjusting by a county random effect estimated from Medicare and Medicaid claims. If that is so, then a county’s racial and age composition moves its published estimate *by construction*, and any regression of these estimates on a county-level social index that contains a demographic component will partly recover the post-stratification machinery rather than a local epidemiologic signal. The Social Vulnerability Index contains exactly such a component (Theme 3, racial and ethnic minority status). We test how much of the apparent social gradient is this artifact.

(2) How we tested it. We obtained VEHSS county estimates separately for each race/ethnicity by age stratum (non-Hispanic White, non-Hispanic Black, and Hispanic adults, each aged 40–64 and 65–84). Within a single stratum, demographic composition is constant by construction, so the only county-varying component is the county random effect, which is the genuinely local signal. We estimated the SVI coefficient (a) on the composition-varying “All races, all ages” file, as a conventional analysis would, and (b) within each fixed stratum, and compared them.

(3) What we found. The composition-varying estimate was $\beta_{\text{SVI}} = 0.146$. The within-stratum estimates were roughly one fifth as large and remarkably consistent across strata:

Specification	β_{SVI}
All races, all ages (composition varies)	0.146
Non-Hispanic White 65–84	0.033
Non-Hispanic Black 65–84	0.033
Hispanic 65–84	0.035
Non-Hispanic White 40–64	0.035
Pooled, stratum fixed effects	0.035

Approximately 76% of the apparent gradient is attributable to demographic composition. Overdispersion also fell sharply, from a dispersion parameter of about 15 in the unstratified model to 1.93 in the within-stratum model, indicating that much of the excess county-level variance was itself compositional.

(4) Interpretation. This is a general caution for the use of VEHSS county estimates, not a finding about diabetic retinopathy. Studies relating these estimates to composite social indices, area deprivation measures, or racial composition will substantially overstate local gradients unless demographic composition is held fixed. Notably, the artifact is specific to demographic predictors: VEHSS does not use poverty or other socioeconomic variables as model inputs, so adjustment for the non-demographic content of SVI is not circular. Within-stratum estimation is a cheap and effective safeguard and we recommend it as routine.

Test 2. Is proximity to academic ophthalmology uniquely protective?

(1) **What is tested.** The motivating hypothesis of the prior version of this work was that counties distant from academic ophthalmology programs face barriers to subspecialty retinal care, and therefore worse diabetic eye outcomes. The identification problem is that academic programs are sited where ophthalmologists already practice. Raw distance to an academic program therefore conflates proximity to academic tertiary capacity with proximity to eye care in general. We test whether anything is attributable to academic centers specifically.

(2) **How we tested it.** Five tests of increasing stringency.

2a. *Conditional effect.* We computed a second exposure, drive time from the population-weighted centroid to the nearest county containing at least one patient-care ophthalmologist, and entered both exposures jointly. The academic coefficient is then the association holding general eye care proximity fixed.

2b. *Discordant counties.* A 2×2 comparison of counties with versus without a local ophthalmologist, crossed with above versus below median drive time to an academic program. This is the comparison a reviewer of the prior submission requested directly.

2c. *Effect modification.* An interaction between academic drive time and having no local ophthalmologist. If academic centers supply something distinctive, their reach should matter most where no local alternative exists.

2d. *Capacity rather than proximity.* A gravity-style access score summing program capacity (ophthalmology residents) discounted by a Gaussian decay in drive time with a 120-minute bandwidth, replacing nearest-program distance.

2e. *Program type.* Distance to the nearest university-based program and to the nearest community-hospital program entered jointly, classified from program names (98 university-based, 25 community/hospital). If the mechanism is subspecialty depth rather than the presence of trainees, university proximity should dominate.

(3) **What we found.**

2a. Academic drive time: -0.024 alone, -0.023 conditional on supply distance. Distance to any ophthalmologist: $+0.029$ alone, $+0.014$ conditional ($P = 0.06$). The two exposures correlate only 0.41. The signs are opposite: distance to *any* ophthalmologist behaves as an access hypothesis predicts, while distance to an *academic* program does the reverse.

2b. The 2×2 :

	<i>n</i>	VTDR share	Median academic drive (min)
Local ophthalmologist, far from academic	396	18.6%	183
Local ophthalmologist, near academic	773	19.6%	61
No local ophthalmologist, far from academic	1,163	16.9%	195
No local ophthalmologist, near academic	787	17.2%	87

Counties with ample local ophthalmology but far from an academic center do not show worse case mix than those near one (18.6% vs 19.6%). The larger difference is horizontal: having any local ophthalmologist is associated with a roughly two point *higher* vision-threatening share.

2c. Interaction $\beta = 0.008$, $P = 0.70$. A clean null.

2d. Capacity-weighted access $\beta = +0.010$, $P = 0.06$, wrong-signed.

2e. University program distance $\beta = -0.028$ ($P < 0.001$); community-hospital program distance $\beta = +0.000$ ($P = 0.99$). The two distances correlate 0.61, so this is a real distinction rather than collinearity. Note this test runs on 1,333 counties, those having a community-hospital program among their routed candidates, a subsample skewed urban and eastern.

In the spatial model reported in the main text, academic drive time retains a small negative posterior mean (-0.010 , 95% CrI -0.019 to -0.001): still signed opposite to the access hypothesis.

(4) Interpretation. No specification supports the hypothesis, including the two most favorable to it (capacity weighting and the no-local-alternative subgroup). The consistent pattern is that proximity to specialist eye care, and especially to university programs where subspecialty retinal diagnosis concentrates, predicts *more* diagnosed vision-threatening disease. This is difficult to read as academic centers improving outcomes and easier to read as diagnostic intensity: where the subspecialists are, vision-threatening disease is more completely identified and coded. The authors of the underlying surveillance estimates anticipated exactly this, cautioning that claims diagnoses are influenced by access to ophthalmologists and that their model may mistake such factors for true variation in prevalence. We regard test 2e as the closest thing in these data to a direct measurement of that phenomenon, while noting that it is an interpretation we have not independently isolated.

Test 3. Does the social gradient operate through insurance coverage?

(1) What is tested. If the social gradient in disease progression is produced by uninsurance and its downstream effects on screening and follow-up, then it should be substantially attenuated in a population with near-universal coverage. Medicare eligibility at age 65 provides that contrast. This also converts a reviewer objection to the prior submission into a design: the reviewer argued that restricting to older adults “enriches the population more likely to have health insurance,” which is precisely what makes the comparison informative.

(2) How we tested it. We estimated the SVI coefficient separately for adults aged 40–64 and 65–84, holding race fixed, with identical covariate sets, and tested the difference with a two-sample z statistic on the coefficient difference.

(3) What we found.

Age band	Mean VTDR share	β_{SVI}	SE
40–64 years	20.3%	0.150	0.0039
65–84 years	18.0%	0.142	0.0040

Difference 0.008; $z = 1.42$; $P = 0.16$; ratio of odds ratios 1.01. The same pattern holds within each race stratum (main text, Table 2): 0.034 versus 0.032 among non-Hispanic White adults, 0.037 versus 0.035 among non-Hispanic Black adults, 0.046 versus 0.037 among Hispanic adults.

(4) Interpretation. Near-universal insurance coverage does not narrow the social gradient in how advanced diabetic eye disease is. Whatever produces this gradient is therefore not principally a coverage mechanism. Remaining candidates include transportation and appointment access, competing demands on patient time, differential dilated examination rates within insured populations, and continuity of the primary care relationship through which retinal screening is ordered. We note the Hispanic 40–64 versus 65–84 contrast (0.046 vs 0.037) is the largest of the three and is the one comparison where a coverage effect cannot be excluded.

Test 4. Which dimension of social vulnerability?

(1) What is tested. SVI decomposes into four themes: socioeconomic status, household characteristics, racial and ethnic minority status, and housing type and transportation. Theme 4 contains a no-vehicle-available component and is the closest thing in the index to a transport access measure. If progression tracked Theme 4 more than Theme 1, that would be an access finding the geographic distance measures failed to capture, and it would be directly actionable.

(2) How we tested it. Each theme was entered separately, and then all four jointly, with the standard covariate set.

(3) What we found. Entered separately, the racial and ethnic minority theme dominated ($\beta = 0.246$) against socioeconomic status (0.135), housing and transportation (0.096), and household characteristics (0.080). Entered jointly, minority status was 0.225 while housing and transportation turned slightly negative (-0.026).

(4) Interpretation. **We do not report this as a substantive finding, and it should not be cited as one.** These models were estimated on composition-varying estimates, and Test 1 establishes that county racial composition moves the VEHSS outcome mechanically. The apparent dominance of the minority status theme is therefore largely the post-stratification artifact reappearing, since Theme 3 is essentially a measure of county racial composition. A properly stratified decomposition would require SVI themes to vary within race strata, which they do only weakly. We report the result here solely to document that we ran it and to explain why it cannot be interpreted. What can be said is the negative point: there is no evidence that the transportation dimension drives the gradient, which is consistent with the null geographic access results in Test 2.

Test 5. Does the gradient survive adjustment for detection?

(1) What is tested. Because the outcome is built from diagnosed disease in claims, the severity share could rise wherever mild disease goes undetected, independent of true progression. If the social

gradient were simply a detection gradient, it should attenuate substantially once detection proxies are included.

(2) How we tested it. We compared the SVI coefficient with no adjustment against the coefficient after adding ophthalmologist and optometrist density, an indicator for having no local ophthalmologist, drive time to the nearest ophthalmologist, drive time to the nearest academic program, population density, and rurality.

(3) What we found. SVI alone 0.163; with detection proxies 0.145, an attenuation of approximately 11%. In the within-stratum specification the coefficient is 0.035 with the same proxies included.

(4) Interpretation. The gradient is not primarily a detection artifact under the proxies available to us, but this test is weak evidence. Provider supply is an imperfect proxy for screening intensity, and the VEHSS county random effects were themselves already adjusted for ophthalmologists per capita, so our adjustment is partly a second pass at a factor already partially removed. Residual detection bias remains the principal threat to the main finding, and its likely direction is to overstate it.

Test 6. Spatial dependence

(1) What is tested. County health outcomes are spatially clustered. If unmeasured regional factors such as state Medicaid policy, regional diabetes management norms, or health system catchments drive both social vulnerability and disease progression, an aspatial model will overstate precision and may bias coefficients.

(2) How we tested it. Moran's I was computed on Pearson residuals using a queen contiguity weights matrix over 2021 TIGER/Line county boundaries. Three island counties had no queen neighbor; each was linked to its nearest county by centroid distance so the adjacency graph was connected, a decision recorded here rather than left implicit. We then refitted the largest single stratum (non-Hispanic White, 65–84, 3,106 counties) with a Leroux conditional autoregressive random effect using CARBayes, with binomial likelihood, VTDR cases as successes and DR cases as trials, 20,000 burn-in and 140,000 iterations thinned by 20.

(3) What we found. Moran's $I = 0.237$ ($P < 0.001$), confirming substantial residual clustering. Estimated spatial dependence was strong ($\rho = 0.82$, 95% CrI 0.73 to 0.91). The social vulnerability coefficient attenuated from 0.032 to 0.021 but its credible interval excluded zero (0.012 to 0.029). Academic drive time was -0.010 (95% CrI -0.019 to -0.001), marginally excluding zero but signed opposite to the access hypothesis. Drive time to the nearest ophthalmologist, ophthalmologist density, and population density all had credible intervals spanning zero.

(4) Interpretation. Roughly a third of the within-stratum social gradient is attributable to broad spatial structure, which is consistent with regional policy and health system factors contributing to it. What remains is a genuinely local association that is not explained by demographic

composition, provider supply, rurality, or spatial position. The contrast between the two exposures under the same spatial model is the cleanest single statement of the paper’s result: social vulnerability predicts more advanced disease, whereas distance from academic ophthalmology predicts marginally *less* advanced disease, which is not the access effect the prior framing proposed.

Test 7. Sensitivity analyses

(1) What is tested. Whether the main result depends on specific analytic choices: the age handling criticized in review, the choice of severity share over disease level, and the possibility that any ratio constructed from VEHSS estimates would show these patterns.

(2) How we tested it. *6a. Age standardization.* We computed an age-standardized severity share weighting stratum-specific shares by the national distribution of DR cases across VEHSS age strata, and compared it with the crude all-ages share.

6b. Level versus severity. We refitted the identical specifications with diagnosed DR prevalence among adults with diabetes as the outcome, to compare against the severity share.

6c. Parallel construction on a different disease. Glaucoma carries an analogous severity split in VEHSS (vision-affecting versus all glaucoma). We built the identical ratio and refitted, asking whether these patterns are specific to diabetic retinopathy or general to VEHSS-derived severity ratios.

(3) What we found. *6a.* Crude and age-standardized shares correlate at $r = 0.992$. Results are unchanged.

6b. With the proper diabetes denominator and full covariate adjustment, drive time was not associated with DR level ($\beta = -0.005$, $P = 0.059$), and SVI was associated only weakly ($\beta = 0.025$) compared with its association with severity ($\beta = 0.145$ unstratified). Disease occurrence and disease progression behave differently, and the severity share is essentially uncorrelated with the level (Spearman $\rho = -0.08$).

6c. The glaucoma severity share has roughly one tenth the relative variation of the VTDR share (coefficient of variation 0.024 vs 0.231) and correlates with it at 0.55.

(4) Interpretation. *6a* retires the age-cutoff objection: the result does not depend on where an age threshold is placed, because no threshold is used and standardization does not move it.

6b is the substantive justification for the paper’s reframing. The social gradient in progression is several times the gradient in occurrence among people with diabetes, and the two outcomes are nearly orthogonal, so this is not the established prevalence finding restated.

6c must be reported carefully. We originally proposed glaucoma as a negative control. It cannot serve that purpose: its severity share barely varies and it correlates substantially with the DR severity share, so a null there would reflect low variance rather than specificity. The shared variation is itself suggestive of a common county-level factor, plausibly care intensity, which is consistent with the detection interpretation in Test 2 but does not establish it. A stronger design

would use a placebo *exposure*, such as distance to accredited programs in a specialty unrelated to eye care, which we did not have data to construct.

Test 8. Detection stress test using Medicare claims volume

(1) What is tested. The principal threat to the main finding is that mild diabetic retinopathy goes undiagnosed more often in disadvantaged counties, shrinking the denominator and inflating the vision-threatening share without any true difference in progression. Provider density (Test 5) is a weak proxy for this, because having ophthalmologists nearby is not the same as diagnostic activity occurring. We wanted a measure of how much diagnostic work actually happens in a county, drawn from claims rather than from workforce counts.

(2) How we tested it. From the CMS Medicare Geographic Variation Public Use File (county, 2021) we took imaging events, evaluation and management events, and diagnostic test events per 1,000 Original Medicare beneficiaries. These were added to the within-stratum specification, singly and jointly, on a common complete-case sample (13,035 cells).

(3) What we found.

Specification	β_{SVI}	SE	P
Base (same sample)	0.0353	0.0016	< 0.001
+ Medicare imaging per 1,000	0.0371	0.0016	< 0.001
+ Medicare E&M per 1,000	0.0375	0.0016	< 0.001
+ all three	0.0354	0.0016	< 0.001

The coefficients on the detection proxies themselves were imaging -0.028 ($P < 0.001$), E&M $+0.010$ ($P < 0.001$), and tests $+0.007$ ($P = 0.002$).

(4) Interpretation. The social gradient is completely unmoved. What makes this informative rather than merely null is that the detection pathway is plainly visible in the same model: counties with more Medicare imaging have a *lower* vision-threatening share, exactly as differential detection predicts, since more diagnostic activity finds more mild disease. So the mechanism exists, is measurable here, and does not account for the social gradient. This is the strongest evidence we can offer against the detection explanation. Its limit is that these are all-cause diagnostic volumes, not retinal screening specifically; a claims measure of dilated eye examination rates by county would be a better instrument and we could not obtain one at county level.

Test 9. Preventive-screening engagement as a mechanism

(1) What is tested. If the gradient is not coverage and not detection, the remaining candidate on our causal diagram is screening cadence and retention in care. We wanted a measure of whether

a county’s population actually completes recommended preventive screening, with no ophthalmic content, so that any attenuation could not be attributed to eye care access.

(2) How we tested it. County mammography screening completion (County Health Rankings 2021, derived from Medicare claims) and influenza vaccination were added to the within-stratum specification.

(3) What we found. The gradient fell from 0.0353 to 0.0206, an attenuation of approximately 42% on the log-odds scale and 29% on the standardized scale. Mammography completion was itself strongly associated with the outcome ($\beta = -0.039$, $P < 0.001$), and correlated -0.42 with social vulnerability. Influenza vaccination added little ($\beta = 0.005$, $P = 0.04$).

(4) Interpretation. Roughly two fifths of the social gradient in diabetic retinopathy progression is shared with general preventive-screening follow-through. Read alongside Test 8, where diagnostic volume explained none of it, the pattern is specific: what matters is not how much medical looking occurs in a county but whether its population completes recommended preventive care. That points at screening cadence rather than at ophthalmology capacity, and it is consistent with the null geographic access results in Test 2.

Two cautions. Mammography is a female-only screen applied here to a both-sex outcome, so it functions as a county-level index of screening culture, not an individual-level mediator. And because screening engagement is plausibly on the causal path rather than a confounder, this is a decomposition, not an adjustment; we do not claim a formal mediation estimate.

Test 10. Direct coverage measures

(1) What is tested. Test 3 inferred the role of insurance indirectly, from the age contrast at Medicare eligibility. Here we test coverage directly.

(2) How we tested it. The county uninsured rate (County Health Rankings) and the share of Medicare beneficiaries dually eligible for Medicaid (CMS) were added to the within-stratum specification. Dual eligibility is informative because within the 65 and over population everyone already has Medicare, so it isolates poverty among the fully insured.

(3) What we found. The gradient moved from 0.0353 to 0.0343 with both included, and neither covariate was independently associated with the outcome (uninsured $P = 0.14$; dual-eligible $P = 0.44$). Restricting to the 65–84 band and using dual eligibility as the exposure, however, gave $\beta = 0.006$ ($P < 0.001$).

(4) Interpretation. Direct coverage measures confirm what the age contrast implied: insurance status does not explain the social gradient in progression. The 65–84 dual-eligibility result is the useful positive: among a population that is universally insured, poverty still predicts more advanced disease. Coverage is necessary but plainly not sufficient.

Test 11. Medicaid expansion, and a placebo that refutes it

(1) What is tested. Medicaid expansion changed coverage for low-income adults under 65 and left the 65 and over population untouched. If coverage drove the social gradient, the gradient among adults 40–64 should be smaller in expansion states. The 65–84 band is a built-in placebo: expansion cannot have affected it, so any contrast appearing equally there is confounding.

(2) How we tested it. Counties were classified by 2021 state expansion status (Kaiser Family Foundation). Missouri and Oklahoma implemented mid-2021 and were excluded rather than forced into either group. We estimated the SVI gradient in each age-by-expansion cell and fitted the formal triple interaction.

(3) What we found.

Age band	Expansion status	Cells	β_{SVI}
40–64	Non-expansion	2,736	0.0163
40–64	Expansion	3,822	0.0461
65–84	Non-expansion	2,436	0.0122
65–84	Expansion	3,307	0.0421

Triple interaction SVI $\times$ expansion $\times$ age band: $\beta = 0.0023$, $P = 0.73$.

(4) Interpretation. Read carelessly, the first two rows say Medicaid expansion triples the social gradient, which would be a striking and counterintuitive policy finding. The placebo refutes it. The same contrast appears, essentially unchanged, in the Medicare-age population that expansion could not have touched, and the triple interaction is null. The difference is therefore between expansion and non-expansion *states*, which differ in region, urbanisation, health system structure, and the distribution of social vulnerability itself, and not a consequence of expansion. We report this test principally as a demonstration that the placebo design works, and as a caution against the two-cell comparison that a reader might otherwise draw from these data.

Test 12. Convergent validity against an independent care-quality outcome

(1) What is tested. Whether the social gradient we report sits within a coherent pattern of county care-quality gradients, or is anomalous and therefore more likely to be a measurement artifact of VEHSS.

(2) How we tested it. We regressed two outcomes drawn entirely outside VEHSS on social vulnerability with the same covariate set: preventable hospital stays (ambulatory-care-sensitive admissions, Medicare claims) and mammography screening completion. All outcomes were placed in standard deviation units so magnitudes are comparable with the vision-threatening share.

(3) What we found. Per standard deviation of social vulnerability: preventable hospital stays +0.279 SD ($P < 0.001$); mammography completion -0.465 SD ($P < 0.001$); vision-threatening

share +0.100 SD ($P < 0.001$). The Spearman correlation between preventable hospital stays and the vision-threatening share was 0.134.

(4) Interpretation. Social vulnerability predicts ambulatory care failure in a different organ system and a different data system, which supports the care-quality reading of our result. It also puts the finding in proportion: the diabetic retinopathy progression gradient is roughly a third the magnitude of the preventable-hospitalisation gradient and about a fifth of the mammography gradient. It is real and consistent, but it is a modest gradient, and the low correlation between the two outcomes indicates they are related rather than interchangeable.

Test 13. Data provenance and reproducibility

(1) What is tested. Whether the analytic file can be rebuilt from public sources without manual steps, and whether derived quantities reproduce published benchmarks.

(2) How we tested it. The pipeline is a numbered sequence of scripts that downloads each source directly from its public endpoint, with no manual download or point-and-click step. We validated the constructed outcome against the published national estimates.

(3) What we found. Aggregating our county numerators reproduces the published national figures: DR 26.43% of adults with diabetes (published 26.43%), VTDR 5.05% (published 5.06%), VTDR share 19.1% (published 1.84/9.60 million = 19.2%). Source coverage was complete except as noted: Puerto Rico is absent from VEHSS; 14 island and remote counties in Alaska and Hawaii have no routable road path to any program.

(4) Interpretation. Two data-handling issues discovered during construction are worth recording because they would silently corrupt a replication. First, the earlier working dataset for this project used 2023 American Community Survey population, whose Connecticut planning-region geography does not match the county geography of the 2021 surveillance and vulnerability data; this produced eight apparently missing Connecticut counties that are not missing at all once the correct vintage is used. Second, one residency program (University of Wisconsin, Madison) geocoded to Madison County, Illinois, roughly 350 miles away, which would have removed Wisconsin's flagship program from the exposure surface for the upper Midwest. It was caught by cross-checking program locations against Area Health Resources File ophthalmology resident counts, a check we recommend to anyone geocoding program rosters.